

1 Question (10 points)

Give context-free grammar that generate the following language. The alphabet is $\{0, 1\}$.

$A = \{w \mid \text{starts with 0 and has odd length, or starts with 1 and has even length}\}$

2 Question (20 points)

Are the following languages context-free? Prove your answer.

- $B = \{0^n 1^n 0^n 1^n | n \geq 0\}$
- $C = \{0^n \# 0^{2n} \# 0^{3n} | n \geq 0\}$

3 Question (10 points)

Give PDA that generate the following language. The alphabet is $\{0, 1\}$.

$D = \{w00w^R \text{ or } w11w^R\}$

4 Question (20 points)

Are the following languages context-free? Prove your answer.

- $E = \{a^n b^m c^m d^n e^p f^p \mid n, m, p \geq 0\}$
- $F = \{a^p b^n c^m d^m e^n f^p \mid n, m, p \geq 0\}$